

Curriculum Vitae

Academic Name: Ayesha Asloob Qureshi

Official name: Ayesha Asloob Topaçoğlu

Research Interests:

My research interest lies in the intersection of Commutative Algebra and Combinatorics.

Combinatorial Commutative Algebra (monomial and binomial ideals arising from posets, hypergraphs, simplicial complexes, polymatroids and polyominoes, toric algebras and combinatorics of affine semigroups)

Homological methods in Commutative Algebra (free resolutions, Betti numbers, regularity, asymptotic homological properties of powers of graded ideals, Cohen-Macaulay and Gorenstein modules)

Gröbner basis theory and applications.

Current Position:

Faculty Member, Sabancı University, Istanbul, Türkiye (September 2016–present)

Previous Positions:

- March 2016- August 2016. Postdoctoral Fellow, The Abdus Salam International Centre of Theoretical Physics, Trieste, Italy.
- December 2014 – February 2016. Postdoctoral Fellow, **Japan Society for the Promotion of Science (JSPS) fellowship**, Osaka University, Japan.
- 2014. Postdoctoral Fellow, The Abdus Salam International Centre of Theoretical Physics, Trieste, Italy.
- 2013. Long-Term Visiting Scientist, The Abdus Salam International Centre of Theoretical Physics, Trieste, Italy.

Education:

- PhD in Mathematics, Abdus Salam School of Mathematical Sciences, GCU, Lahore, Pakistan, 2007-2012.

Thesis Title: **Ideals generated by quadratic binomials, Hibi relations and Polyominoes**

Advisor: **Prof. Dr. Jürgen Herzog**, (Universität Duisburg-Essen, Germany)

- Master of Science (MSc) in Mathematics, Government College University, Lahore, Pakistan, 2005–2007.

Awards:

- **BAGEP Award, The Science Academy Young Scientist Award, Türkiye, 2024.**
- **Abdus Salam Medal (for Young Researchers),** National Mathematical Society of Pakistan (NMSP), 2018.
- **Best student award** in second year of PhD, Abdus Salam School of Mathematical Sciences, Lahore, Pakistan, 2010.
- **Best student award** in first year of PhD, Abdus Salam School of Mathematical Sciences, Lahore, Pakistan, 2009.
- **First position with Gold Medal** in MSc (Mathematics) from Government College University, Lahore, Pakistan, 2007.
- **Third position** in BSc from Government College University, Lahore, Pakistan, 2005.

Selected Fellowships/Research visits:

- **September 2026:** Research in Pairs Program, Mathematisches Forschungsinstitut Oberwolfach (MFO), Germany.
- **Junior Associate Fellowship** at The Abdus Salam International Centre of Theoretical Physics, Trieste, Italy, 2019–2024.
- **PNPD–CAPES Postdoctoral Fellowship**, Instituto de Ciências Matemáticas e de Computação, São Carlos, Brazil (May 2016; declined).

Grants/ Projects

- **TÜBİTAK 1001 (2024-2027)**
Title: “*Kombinatorikten Ortaya Çıkan Çift Terimli İdeallerin Kuvvetleri*”
Role: Principal Investigator
- **TÜBİTAK 1001 (2022-2025)**
Title: “*Karesiz Tekterimli İdeallerin Kuvvetlerinin Asimptotik Özellikleri*”
Role: Principal Investigator.
- **TÜBİTAK 2509 (2019-2021)** Franse Dışişleri Bakanlığı ile pia Bosphorus İşbirliği Programı
Title: “*Gröbner Bases, Resultants and Polyhedral Geometry - GRAPE*”,
Role: Researcher
- **TÜBİTAK 3501(2019-2022)** Kariyer geliştirme Programı
Title: “*Poliomino ideallerinin cebirsel ve homolojik özellikleri*”
Role: Principal Investigator
- “2019- Erasmus+ Staff Mobility for teaching” to visit Ovidius University of Constanta, Romania. (Declined due to Pandemic in 2020)
- “2017- Erasmus+ Staff Mobility for teaching” to visit Ovidius University of Constanta, Romania.

Invited Talks at Workshops, Schools, and Conferences (last 10 years):

1. July 8–10, 2026: *II Anatolian Algebra Workshop*, Izmir Institute of Technology, Izmir, Türkiye.
2. June 25–26, 2026: *Commutative Algebra, Geometry and Computational Aspects*, Accademia Peloritana dei Pericolanti, Messina, Italy.
3. November 28, 2025: Online: *Combinatorial Commutative Algebra in Canada* (CCAC) Seminar series, Canada.
4. September 8, 2025: *One Day Workshop on Commutative Algebra and Related Fields*, University of Ovidius, Constanta, Romania.
5. August 5–6, 2025: *Combinatorics and Geometry in Istanbul*, Bahçeşehir University, Istanbul.
6. July 21–25, 2025: *International Conference on Commutative Algebra to the Memory of Jürgen Herzog*, VIASM, Hanoi, Vietnam.
7. April 17–23, 2025: *2025 AESIM school on Combinatorial Commutative Algebra: Theory, Computation, and Applications*, COMSATS, Lahore, Pakistan.
8. September 8–13, 2024: *Research School on Topology and Algebra of Singularities and their Applications*, University of Ovidius, Constanta, Romania.
9. June 28–30, 2023: *Genç Matematikçiler Toplantısı*, Turkish Mathematical Society, Nesin Mathematics Village, Sirince, Türkiye.
10. March 19–25, 2023: *CoCoA 2023 International school on Computational commutative algebra*, Hue, Vietnam.
11. May 1–12, 2023: *Workshop on commutative algebra and algebraic geometry in prime characteristic*, International Centre of Theoretical Physics, Trieste, Italy.
12. August 30–September 12, 2021: *Matematik Araştırma Programı (MAP)*, İstanbul Matematiksel Bilimler Merkezi, Boğaziçi Üniversitesi, İstanbul, Türkiye.
13. August 5, 2021: Online: *Commutative Algebra Webinars* organized by the Commutative Algebra Group of IPM, Iran.
14. June 25–26, 2021: Online: *The first Asian CWM ambassadors meeting* organized by CWM-IMU.
15. May 1–2, 2021: Online: *9. Buluşma Bahar Matematik Buluşması*. Türkiye.
16. September 1, 2020: Online, *One day joyful workshop on Commutative Algebra and related fields* organized by University of Ovidius, Romania.
17. September 28, 2019: *One day joyful workshop on Commutative Algebra and related fields* organized by University of Ovidius, Romania.
18. July 29–August 1, 2019: *The 8th SEAMS-UGM International Conference on Mathematics and Its Applications*, Yogyakarta, Indonesia.
19. July 30, 2019: *South East Asian Women Mathematicians Meeting (SEAWM2)*, Yogyakarta, Indonesia.
20. August 26–September 1, 2018: *The 26th National school on Algebra*, Constanta, Romania.
21. May 16–20, 2018: *Antalya Algebra Days XX*, Nesin Mathematics Village, Sirince, Türkiye.
22. December 26–29, 2017: *First winter workshop on advanced topics in Mathematics: Commutative Algebra and Banach space theory*, Abdus Salam School of Mathematical Sciences, Lahore, Pakistan.
23. August 15–19, 2017: *Workshop on contemporary topics in Analysis, Algebra and Geometry*, Abdus Salam School of Mathematical Sciences, Lahore, Pakistan.

24. March 4, 2017: *Ankara-İstanbul Algebraic Geometry & Number Theory meetings*, Istanbul Center for Mathematical Sciences (IMBM), Istanbul.
25. December 9–10, 2016: *CASM Workshop on Algebra and Applications*, Lahore University of Management Sciences (LUMS), Lahore, Pakistan.
26. November 2–3, 2016: *Second International workshop on modern aspects of algebra and graph theory*, COMSATS Institute of Information Technology, Lahore, Pakistan.
27. August 17–24, 2016: *Multigraded algebra and applications, The 24th National School on Algebra*, Moieciu de Sus, Romania.

Postdoctoral and Graduate Supervision:

Postdoctoral Researchers:

Dharm Veer (May–August, 2026) Sabancı University
 Francesco Navarra, (2023–present) Sabancı University
 Marie Amalore Nambi, (2025–present) Sabancı University

Masters Students:

Aslı Tuğcuoğlu Musapaşaoğlu (2018) Sabancı University
 Büşra Atar (2021) Sabancı University
 Elshani Kamberi (2021) Sabancı University
 Erdem Safak Öztürk (2024) Sabancı University

PhD:

Rizwan Jahangir (2024), Sabancı University
 Aslı Tuğcuoğlu Musapaşaoğlu (2024) Sabancı University
 Elshani Kamberi (2025) Sabancı University

Yoshua Yonatan (2023, co-advisor) Institut Teknologi Bandung, Indonesia

Erdem Safak Öztürk (ongoing), Sabancı University

Teaching Experience:

Undergraduate Level Courses: Calculus 1, Calculus 2, Linear Algebra, Discrete Mathematics, Elementary Number Theory, Introduction to Combinatorics, Introduction to Algebra

Graduate Level Courses: Algebra 1, Algebra 2,
 Selected topics in Algebra: Homological Algebra,
 Selected topics in Algebra: Commutative Algebra,
 Selected topics in Algebra: Graded Syzygies,
 Commutative Algebra 1, Commutative Algebra 2

University Committee membership:

2025-2026 Mathematics Seminar coordinator

2017-2025 Coordinator/Member of Research Awards Committee

2017-2018 Graduate Coordinator

Visiting researchers: (Invited researchers at Sabanci University for research collaboration in period 2017-2025)

Jürgen Herzog (Universität of Duisberg-Essen, Germany) – 5 visits

Takayuki Hibi (Osaka University, Japan)

Naoki Terai (Okayama University, Japan)

Giancarlo Rinaldo (University of Messina, Italy)

Sara Saeedi Madani (IPM, Iran)

Seyed Amin Seyed Fakhari (Universidad de los Andes, Colombia)

Somayeh Moradi (Universität of Duisberg-Essen , Ilam University, Iran)

Massomeh Rahimbeigi (University of Kurdistan, Sanandaj, Iran) - 2 visits

Dharm Veer (Dalhousie University, Canada) – 2 visits

Francesco Navarra (University of Messina, Italy)

Francesco Romeo (Università degli Studi di Trento, Italy)

Siti Humaira (Institut Teknologi Bandung, Indonesia)

Yoshua Yonatan (Institut Teknologi Bandung, Indonesia) -2 visits

Workshop Organization:

- Member of Scientific Committee of “Commutative Algebra, Geometry and Computational Aspects”, June 25–26, 2026, Accademia Peloritana dei Pericolanti, Messina, Italy.
- Member of Scientific Committee of “28th Iran Algebra Seminar”, July 10–11, 2024, University of Maragheh, Iran.
- Spring Semester, 2017: Co-organizer of Ankara-İstanbul Algebraic Geometry & Number Theory meetings, Istanbul Center for Mathematical Sciences (IMBM), Istanbul.

Editorial Board Membership:

Galois Journal of Algebra

Selected Refereeing Service:

Journal of Algebra

Proceedings of the American Mathematical Society

European Journal of Combinatorics

Journal of the European Mathematical Society

Journal of Combinatorial Theory, Series A

Mathematische Nachrichten

Collectanea Mathematica

Journal of Pure and Applied Algebra

Journal of Algebraic Combinatorics

Communications in Algebra
Journal of Algebra and its Applications
Illinois Journal of Mathematics
International Journal of Algebra and Computation
Georgian Mathematical Journal
Haceteppe Journal of Mathematics and Statistics
Bulletin of the Malaysian Mathematical Sciences Society
Indonesian Journal of Mathematics

Articles in international journals:

1. Antonino Ficarra, Ayesha Asloob Qureshi, "The homological shift algebra of a monomial ideal", accepted in *Collectanea Mathematica*.
2. Takayuki Hibi, Ayesha Asloob Qureshi, Sara Saeedi Madani, "Complementary edge ideals", *Expositiones Mathematicae*, vol. 44, article 125780, 2026. DOI: 10.1016/j.exmath.2026.125780
3. Elshani Kamberi, Francesco Navarra, Ayesha Asloob Qureshi, "On squarefree powers of facet ideals of simplicial trees", *Journal of Algebra*, vol. 693, pp 240-276, 2026. DOI: doi.org/10.1016/j.jalgebra.2026.01.008
4. Jürgen Herzog, Somayeh Moradi, Ayesha Asloob Qureshi, "Toric rings attached to simplicial complexes", *Collectanea Mathematica*, Published online, 2025. DOI: doi.org/10.1007/s13348-025-00493-9
5. Antonino Ficarra, Ayesha Asloob Qureshi, "Edge ideals and their asymptotic syzygies", *Journal of Pure and Applied Algebra*, vol. 229 (10), pp 108079, 2025. DOI: doi.org/10.1016/j.jpaa.2025.108079
6. Ayesha Asloob Qureshi, Asli Musapasaoglu, "Componentwise linearity of dominating ideals of path graphs", *Haceteppe Journal of Mathematics and Statistics*, vol. 54 (4), pp 1345-1355, 2025. DOI: doi.org/10.15672/hujms.1483083
7. Ayesha Asloob Qureshi, Somayeh Bandari, "Componentwise linear ideals and exchange properties", *Journal of Algebra and its Applications*, Published online, 2025. DOI: doi.org/10.1142/S0219498826502014
8. Takayuki Hibi, Jürgen Herzog, Somayeh Moradi, Ayesha Asloob Qureshi, "The divisor class group of a discrete polymatroid", *Journal of Combinatorial Theory Series A*, vol. 205, pp 105869, 2024. DOI: doi.org/10.1016/j.jcta.2024.105869
9. Mehrdad Nasernejad, Ayesha Asloob Qureshi, "Algebraic implications of neighborhood hypergraphs and their transversal hypergraphs", *Communications in Algebra*, vol. 52(6), pp. 2328-2345, 2024. DOI: [10.1080/00927872.2023.2300760](https://doi.org/10.1080/00927872.2023.2300760).
10. Mehrdad Nasernejad, Asli Musapasaoglu, Ayesha Asloob Qureshi, "The edge ideals of t-spread d-partite hypergraphs", *Collectanea Mathematica*, vol. 75, pp. 735-751, 2024.

DOI: [10.1007/s13348-023-00410-y](https://doi.org/10.1007/s13348-023-00410-y).

11. Somayeh Bandari, Ayesha Asloob Qureshi, "Ideals with linear quotients and componentwise polymatroidal ideals", *Mediterranean Journal of Mathematics*, vol. 20, article no: 53, 2023. DOI: [10.1007/s00009-023-02264-x](https://doi.org/10.1007/s00009-023-02264-x).
12. Mehrdad Nasernejad, Ayesha Asloob Qureshi, Kazem Kashyarmansh, Lesli G. Roberts, "Classes of normally and nearly normally torsion-free ideals", *Communications in Algebra*, vol. 50 (9), pp. 3715-3733, 2022. DOI: [10.1080/00927872.2022.2042546](https://doi.org/10.1080/00927872.2022.2042546).
13. Mehrdad Nasernejad, Kazem Kashyarmansh, Ayesha Asloob Qureshi, "On the matroidal path ideals", *Journal of Algebra and Its Applications*, vol. 22(11), pp. 2350227, 2023. DOI: [10.1142/S0219498823502274](https://doi.org/10.1142/S0219498823502274).
14. Ayesha Asloob Qureshi, Giancarlo Rinaldo, Francesco Romeo, "Hilbert series of parallelogram polyominoes", *Research in the Mathematical Sciences*, vol. 9, article no: 28, 2022. DOI: [10.1007/s40687-022-00323-5](https://doi.org/10.1007/s40687-022-00323-5).
15. Mehrdad Nasernejad, Somayeh Bandari, Asli Musapasaoglu, Ayesha Asloob Qureshi, "Dominating ideals and closed neighborhood ideals of graphs", *Mediterranean Journal of Mathematics*, vol. 20, article no: 53, 2023. DOI: [10.1007/s00009-022-02107-1](https://doi.org/10.1007/s00009-022-02107-1).
16. Mirsadegh Sayedsadeghi, Mehrdad Nasernejad, Ayesha Asloob Qureshi, "On the embedded associated primes of monomial ideals", *Rocky Mountain Journal of Mathematics*, vol. 52(1), pp. 275–287, 2022. DOI: [10.1216/rmj.2022.52.275](https://doi.org/10.1216/rmj.2022.52.275).
17. Nursel Erey, Ayesha Asloob Qureshi, "Second power of cover ideals of paths," *Algebra Colloquium*, vol. 29(4), pp. 669–686, 2022. DOI: [10.1142/S1005386722000487](https://doi.org/10.1142/S1005386722000487).
18. Viviana Ene, Jürgen Herzog, Ayesha Asloob Qureshi, Francesco Romeo, "Regularity and Gorenstein property of L-convex polyominoes," *Electronic Journal of Combinatorics*, vol. 28(1), PI.50, 2021. DOI: [10.37236/9531](https://doi.org/10.37236/9531).
19. Hamed Mousavi, Farzad Padashnik, Ayesha Asloob Qureshi, "On reduced Archimedean skew power series rings", *Journal of Algebra and Its Applications*, vol. 21(03), pp. 2250042, 2022. DOI: [10.1142/S0219498822500426](https://doi.org/10.1142/S0219498822500426).
20. Rodica Dinu, Jürgen Herzog, Ayesha Asloob Qureshi, "Restricted classes of Veronese type ideals and algebras", *International Journal of Algebra and Computation*, vol. 31(1), pp. 173-197, 2021. DOI: [10.1142/S0218196721500090](https://doi.org/10.1142/S0218196721500090)
21. Viviana Ene, Jürgen Herzog, Ayesha Asloob Qureshi, "t-spread strongly stable monomial ideals", *Communications in Algebra*, vol. 47(12), pp. 5303-5316, 2019. DOI: [10.1080/00927872.2019.1617876](https://doi.org/10.1080/00927872.2019.1617876)
22. Jürgen Herzog, Ayesha Asloob Qureshi, Maryam Mohammadi Saem, "Fiber cone of a monomial ideal in two variables", *Journal of Symbolic Computation*, vol. 94, pp. 52-69, 2019. DOI: [10.1016/j.jsc.2018.06.022](https://doi.org/10.1016/j.jsc.2018.06.022)
23. Jürgen Herzog, Ayesha Asloob Qureshi, Akihiro Shikama, "On the relations of isotonian algebras", *Proceedings of the American Mathematical Society*, vol. 145(10), pp. 4119-4226, 2017. DOI: [10.1090/proc/13502](https://doi.org/10.1090/proc/13502)

24. Ayesha Asloob Qureshi, Takafumi Shibata, Akihiro Shikama, "Simple polyominoes are prime", *Journal of Commutative Algebra*, vol. 9(3), pp. 413-422, 2017. DOI: [10.1216/JCA-2017-9-3-413](https://doi.org/10.1216/JCA-2017-9-3-413)
25. Mina Bigdeli, Jürgen Herzog, Takayuki Hibi, Akihiro Shikama, Ayesha Asloob Qureshi, "Isotonian Algebras", *Nagoya Mathematical Journal*, vol. 230, pp. 83-101, 2018. DOI: [10.1017/nmj.2017.4](https://doi.org/10.1017/nmj.2017.4)
26. Faryal Chaudhry, Ayesha Asloob Qureshi, "Hankel ideals of 2-minors associated with pairs of proper interval graphs", *Journal of Algebra and Its Applications*, vol. 16(4), pp. 1750070, 2017. DOI: [10.1142/S0219498817500700](https://doi.org/10.1142/S0219498817500700)
27. Jürgen Herzog, Ayesha Asloob Qureshi, Akihiro Shikama, "Alexander duality for monomial ideals associated with isotone maps between posets", *Journal of Algebra and Its Applications*, vol. 14(5), pp.1650089, 2016. DOI: [10.1142/S0219498816500894](https://doi.org/10.1142/S0219498816500894)
28. Jürgen Herzog, Ayesha Asloob Qureshi, Akihiro Shikama, "Gröbner bases of balanced polyominoes", *Mathematische Nachrichten*, vol. 288(7), pp. 775-783, 2015. DOI: [10.1002/mana.201400153](https://doi.org/10.1002/mana.201400153)
29. Jürgen Herzog, Takayuki Hibi, Ayesha Asloob Qureshi, "Polarization of Koszul cycles with applications to powers of edge ideals of whisker graphs", *Proceedings of the American Mathematical Society*, vol. 143(7), pp. 2767-2778, 2015. DOI: [10.1090/S0002-9939-2015-12456-2](https://doi.org/10.1090/S0002-9939-2015-12456-2)
30. Jürgen Herzog, Ayesha Asloob Qureshi, "Persistence and stability properties of powers of ideals", *Journal of Pure and Applied Algebra*, vol. 219(3), pp. 530-542, 2015. DOI: [10.1016/j.jpaa.2014.05.011](https://doi.org/10.1016/j.jpaa.2014.05.011)
31. Ayesha Asloob Qureshi, "Indispensable Hibi relations and Gröbner basis", *Algebra Colloquium*, vol. 22(4), pp. 567-580, 2015. DOI: [10.1142/S1005386715000498](https://doi.org/10.1142/S1005386715000498)
32. Takayuki Hibi, Ayesha Asloob Qureshi, "Nonsimple polyominoes and prime ideals", *Illinois Journal of Mathematics*, vol. 59(2), pp. 391-398, 2015. DOI: [10.1215/ijm/1462450707](https://doi.org/10.1215/ijm/1462450707)
33. Viviana Ene, Jürgen Herzog, Takayuki Hibi, Ayesha Asloob Qureshi, "The binomial edge ideal of a pair of graphs", *Nagoya Mathematical Journal*, vol. 213, pp. 105-125, 2014. DOI: [10.1215/00277630-2389872](https://doi.org/10.1215/00277630-2389872)
34. Sarfraz Ahmad, Imran Anwar, Ayesha Asloob Qureshi, "On characteristic poset and Stanley decomposition", *Analele Ştiinţifice ale Universităţii Ovidius Constanţa - Seria Matematică*, vol. 22(2), pp. 21-28, 2015. DOI: [10.2478/auom-2014-0027](https://doi.org/10.2478/auom-2014-0027)
35. Viviana Ene, Ayesha Asloob Qureshi, Asia Rauf, "Regularity of join-meet ideals of distributive lattices", *Electronic Journal of Combinatorics*, vol. 20(3), article no: P20, 2013. DOI: [10.37236/3621](https://doi.org/10.37236/3621)
36. Viviana Ene, Ayesha Asloob Qureshi, "Ideals generated by diagonal 2-minors", *Communications in Algebra*, vol. 41(8), pp. 3058-3066, 2013. DOI: [10.1080/00927872.2012.672603](https://doi.org/10.1080/00927872.2012.672603)
37. Ayesha Asloob Qureshi, "Ideals generated by 2-minors, collections of cells, and stack polyominoes", *Journal of Algebra*, vol. 357(3), pp. 279-303, 2012. DOI: [10.1016/j.jalgebra.2012.01.032](https://doi.org/10.1016/j.jalgebra.2012.01.032)

Submitted articles:

1. Francesco Navarra, Ayesha Asloob Qureshi, Giancarlo Rinaldo, *Switching rook polynomial of Convex collections of cells*, preprint available at arXiv:2511.22346
2. Francesco Navarra, Ayesha Asloob Qureshi, Giancarlo Rinaldo, *Switching Rook Polynomials of Collections of Cells: Palindromicity and Domino-Stability*, preprint available at arXiv:2511.13982
3. Francesco Navarra, Ayesha Asloob Qureshi, Dharm Veer, *t-Young complexes and squarefree powers of t-path ideals*, preprint available at arXiv:2511.13982
4. Takayuki Hibi, Francesco Navarra, Ayesha Asloob Qureshi, Sara Saeedi Madani, *Minimal primes and radicality of ideals generated by adjacent 2-minors*, preprint available at arXiv:2512.21449
5. Marie Amalore Nambi, Ayesha Asloob Qureshi, *Squarefree powers of closed neighborhood ideals*, preprint available at arXiv:2603.15229
6. Takayuki Hibi, Ayesha Asloob Qureshi, Sara Saeedi Madani, *Prime and Cohen-Macaulay binomial ideals with linear resolutions*, submitted.
7. Takayuki Hibi, Ayesha Asloob Qureshi, *Prime and Cohen--Macaulay ideals of 2-minors with linear resolution*, submitted.
8. Francesco Navarra, Ayesha Asloob Qureshi, Seyed Amin Seyed Fakhari, *An algebraic study of ideals of weak graph homomorphisms*, preprint available at arXiv:2606.26689
9. Francesco Navarra, Ayesha Asloob Qureshi, Naoki Terai, *On the Linearity of Squarefree Powers of Edge Ideals*, preprint available at arXiv:2607.00838